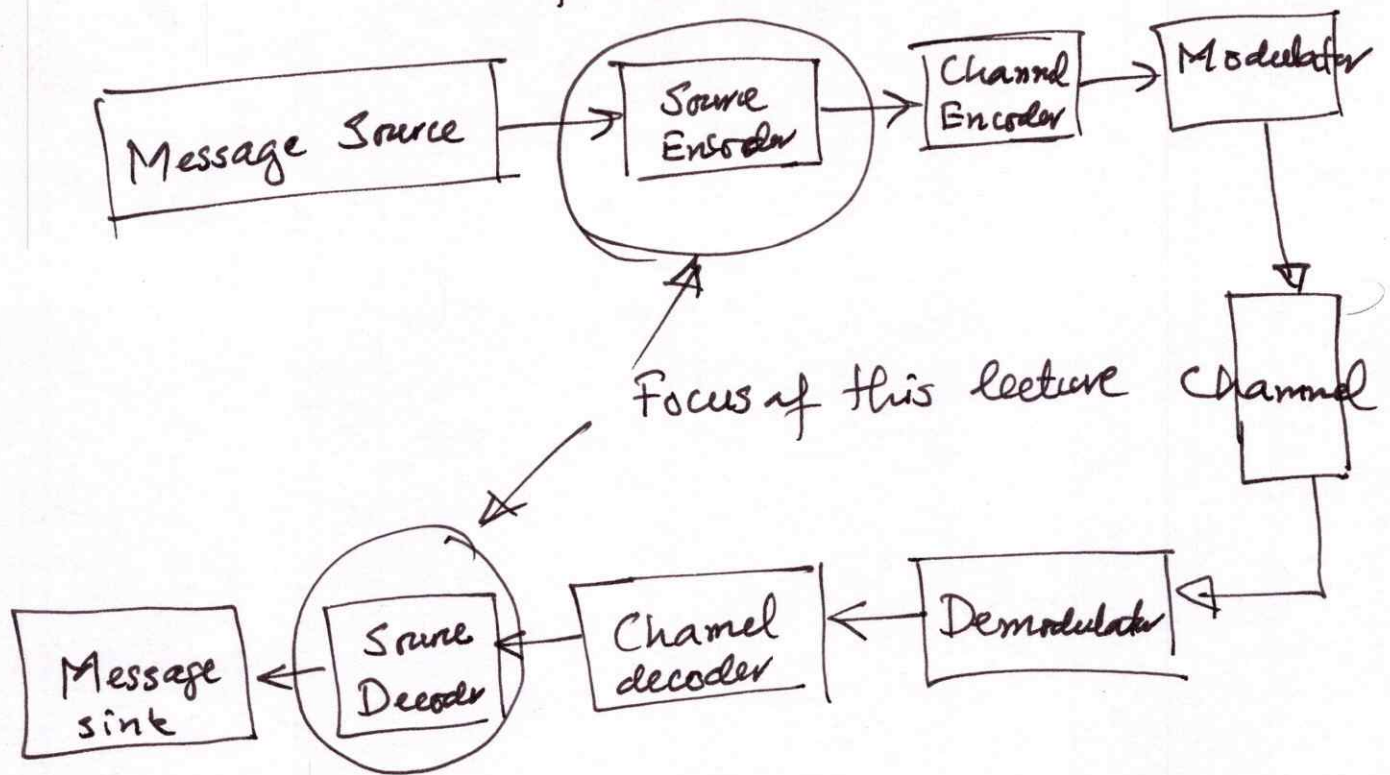


ADC Notes : Chapter 2 (Modern Comm. Systems) (31/9/2025)

Sources of Information.

①



2.1 Major Types of Signals.

Speech & Audio Signals.

— The first type of signals for which comm systems were used were narrowband speech signals.

— Human voice and hearing $\rightarrow 80\text{Hz} - 20\text{kHz}$.

— Smallest freq. band able to convey emotions and speaker identification $\rightarrow 300\text{Hz} - 3.4\text{kHz}$.

Voice band

Low-quality signals,

— However there is wideband speech.

— Telecommunication Union (ITU) standardized wideband audio as G.722.

G.722: frequencies up to 7 kHz sampled at 16 kHz, quantized to 16 bit.

After sampling and quantization the wide band audio signal can be significantly compressed.

— Voice calls NOT over the public switched telephone network (PSTN) can be wideband.

— PC-to-PC calls over skype

Voice digitization happens over software.

(can use G.711 / G.722 / Proprietary voice codec)

— Range of sampling frequencies are supported

8, 12, 16, 24 kHz. (Significantly better than PSTN calls.)

— Cellular telephony^{is} also digital — it supports (3) wideband audio.

— The highest quality audio signals are called entertainment quality.

E.g., CD-audio format

range: 20^{Hz} - 20kHz sampled @ 44.1kHz @ 16-bit resolution.

— The raw data rate is very high

— lossy compression at a high ratio is possible.

lossy compression: high compression ratios that do not allow perfect reconstruction.

— Video: "A video signal is a sequence of still images displayed in rapid succession at a constant rate".

— Video signals project a 3D world onto a 2D plane.

— The rate at which frames are displayed

is called the frame rate and measured in
frames per second (fps).

(4)

For the human there is a minimum frame rate to give the illusion of continuous movement (24 fps) still used in movie film projection.

- Historical frame rate was 30 fps for VCR and DVD videos — Now its 60 fps.

Analog video $\rightarrow$ digital video.

-
- When analog video is sampled, the amplitude is represented by a number.

- Digital video also comprises of still digital images.
- A digital signal is a rectangular grid of picture elements called pixels.
- Assuming width of a frame is w pixels

and frame height is H pixels, the frame size is $W \times H$ pixels.

(5)

— Density is related to spatial & resolution

320 x 240 (240p)

480 x 360 (360p)

640 x 480 (480p)

1920 x 1080 (1080p)

— Pixels have brightness + color.

— There is a certain number of bits per pixel (bpp). The higher the bpp the subtler the color variation.

— Gray-scale images use 8 bits for a total of $2^8 = 256$ Gray levels.

— A widely used approach for color video is 24 bits (8 bit each R, G, B). — the so called absolute color space.

With these notations the average data rate and total video size are

$$\text{data rate} = W \times H \times \text{bpp} \times \text{fps}$$

$$\text{video size} = \text{data rate} \times \text{duration.}$$

- Average data is high
- Instantaneous data rate is even higher.
- reason is blanking period (a period of time without video data) and each frame also has a blanking period. $\Rightarrow$ video stream is bursty.
- high resolution examples:
 - 8K resolution (7680 $\times$ 4320)
 - 4K resolution (3840 $\times$ 2160)
 - HDTV (1080p) (1920 $\times$ 1080).
- HUGE BIT RATES.

Video

(Monday) 10/6/2025 or 6/10/2025

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Compression: video allows compression by a

large factor using lossy compression algorithms. ^{it} _{if the} _{need}

MPEG, JPEG for audio, images, video.

- While lossy they are perceptually lossless to human hearing and visual systems.

The significance of signal types:

Signals impose different requirements on communication systems.

- Due to human perception voice + video require LOW DELAY.
 - E.g., interactive voice requires a delay $\leq 150\text{ms}$
- However voice + video signals have less demands on packet loss.
- The human ear can tolerate greater than 20% PER (packet error rate)

Up to 3%.

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- The human visual system can similarly tolerate a similar number of bit errors.
- On the other hand, data is less sensitive to delay, but tolerates 0% PER.

2-2 Signal Power and Energy.

Several classifications of signals are possible

periodic vs. non-periodic

continuous vs. discrete

continuous-time vs. discrete time

real-valued vs. complex valued.

The inner product of signals $\langle x(t), y(t) \rangle$ ✓
Pme.

ORTHOGONALITY OF SIGNALS.

Two signals are orthogonal iff $\langle u, v \rangle = 0$ (9)

$$E u = \|u\|_2^2 = \int_{\mathbb{R}} |u(t)|^2 dt$$

For example find the energy of the triangular pulse

$$\Lambda(t) = f(t) = \begin{cases} 1-t & 0 \leq t \leq 1 \\ 1+t & -1 \leq t \leq 0. \end{cases}$$

$$\text{We have } E f(t) = \frac{2}{3}.$$

$$\text{Since } \Lambda(t) \text{ is even } E \Lambda(t) = 2 \int_0^1 (1-t)^2 dt.$$

If the signal $x(t)$ is ~~over~~ an electrical signal (voltage or ^{current} power) then

$$\|x\|_2^2 = \int_{-b}^{+b} |x(t)|^2 dt$$

has units of Joules dissipated in $1-\Omega$ resistor.

Power Signals: So far we have seen

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$$E_u < \infty.$$

i.e., energy signals.

Most real signals can be approximated by infinite duration signals since their duration, in general, is much longer than the inverse of the bandwidth. $\rightarrow$ Clearly infinitely long signals are not energy signals.

We introduce here power signals.

- e.g., periodic signals
- e.g., noise or random signals.

Suppose $x(t)$ is infinitely long.

- Truncate it to $[-\frac{T}{2}, \frac{T}{2}]$

$$x_T(t) = \begin{cases} x(t) & -T/2 \leq t \leq T/2 \\ 0 & \text{else} \end{cases}$$

$x_T(t)$ is a snapshot of a segment of $x(t)$ (11)

of duration of T . The inner product is modified as follows-

$$\langle \underline{x}, \underline{y} \rangle = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x(t) y^*(t) dt.$$

If the signals are periodic then the limit is removed.

$$\langle \underline{x}, \underline{y} \rangle = \frac{1}{T} \int_T x(t) y^*(t) dt$$

The inner product of a signal with itself has physical meaning of power

$$P_{\underline{x}} = \langle \underline{x}, \underline{x} \rangle = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |x_T(t)|^2 dt.$$

For a periodic signal

$$P_x = \frac{1}{T} \int_T |x_T(t)|^2 dt$$

Power of a periodic signal:

(12)

Example 2-2:

If $x(t)$ is periodic with period T then show that

$$\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |x(t)|^2 dt = \frac{1}{T} \int_T |x(t)|^2 dt.$$

Solution: Since the signal is periodic:

$$x(t) = x(t + T)$$

$$\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |x(t)|^2 dt$$

$$= \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |x(t+T)|^2 dt$$

$$= \lim_{T \rightarrow \infty} \frac{1}{2(MT+E)} \int_{-MT-E}^{MT+E} |x(t+T)|^2 dt$$

($T = MT + E$): substitution.

$$\lim_{M \rightarrow \infty} \frac{1}{2MT+E} \left[\int_{-MT-E}^{-MT} |x(t+T)|^2 dt \right]$$

$$+ \int_{-MT}^{MT} |x(t+T)|^2 dt + \int_{MT}^{MT+\epsilon} |x(t+T)|^2 dt \quad (13)$$

The integrals from $-MT-\epsilon$ to $-MT$ and MT to $MT+\epsilon$ go to zero as M approaches ∞ . We are thus

left with

$$\begin{aligned} & \lim_{M \rightarrow \infty} \frac{1}{2(MT+\epsilon)} \int_{-MT}^{MT} |x(t+T)|^2 dt \\ &= \lim_{M \rightarrow \infty} \frac{1}{2(MT+\epsilon)} \sum_{k=-M}^{M-1} \int_{kT}^{(k+1)T} |x(t+T)|^2 dt \end{aligned}$$

Select an arbitrary t_0 and using the change of variables

$t' = t + t_0 - kT$, we obtain:

$$= \lim_{M \rightarrow \infty} \frac{1}{2(MT+\epsilon)} \sum_{k=-M}^{M-1} \int_{t_0}^{t_0+T} |x(t' + \cancel{kT})|^2 dt'$$

$$(t = t' - t_0 + kT) = \lim_{M \rightarrow \infty} \frac{1}{2(MT+\epsilon)} \sum_{k=-M}^{M-1} \int_T |x(t')|^2 dt' \quad \text{constant.}$$

$t+T$

$$t = kT \Rightarrow t' = t_0$$

$$t = (k+1)T \Rightarrow t' = t_0 + T$$

$$= \lim_{M \rightarrow \infty} \left(\frac{1}{2(MT+\epsilon)} ((M-1) - (-M) + 1) \right) \times \int_T |x(t)|^2 dt'$$

$$= \lim_{M \rightarrow \infty} \frac{2M}{2(MT + \epsilon)} \int_T |x(t)|^2 dt$$

$$= \frac{1}{T} \int_T |x(t)|^2 dt.$$

A periodic signal is a power signal if its energy over a single period is finite.

Examples of power signals:

The trigonometric functions:

$$\cos\left(\frac{2\pi n t}{T}\right); \quad n \in \mathbb{Z}$$

$$\sin\left(\frac{2\pi n t}{T}\right); \quad n \in \mathbb{Z}.$$

Proof: The inner products,

$$\frac{1}{T} \int_T \cos\left(\frac{2\pi n t}{T}\right) \cos\left(\frac{2\pi k t}{T}\right) dt = \begin{cases} 0, & n \neq k \\ 1/2, & n = k \neq 0 \\ 1, & n = k = 0, \end{cases}$$

Same for $\frac{1}{T} \int_T \sin\left(\frac{2\pi n t}{T}\right) \sin\left(\frac{2\pi k t}{T}\right) dt$

and finally $\frac{1}{T} \int \cos\left(\frac{2\pi n t}{T}\right) \sin\left(\frac{2\pi k t}{T}\right) dt = 0$

They orthogonality over the time periods that are complete intervals.

These properties are essentially encoded in the orthogonality properties of the complex exponentials:

$$\frac{1}{T} \int e^{j \frac{2\pi n t}{T}} e^{-j \frac{2\pi k t}{T}} dt = \delta_{nk} \quad (\text{Kronecker Delta}).$$

$$\delta_{nk} = \begin{cases} 1 & n = k \\ 0 & n \neq k. \end{cases}$$

HW 2. Derive these results (4 results).

Units of Energy $\rightarrow$ Joules

Units of Power $\rightarrow$ Watts.

For power signals $\|x(t)\|^2 \rightarrow$ Power in the signal

For energy signal $\|x(t)\|^2 \rightarrow$ Energy in the signal.

If $x(t)$ is a voltage signal, then Eq. (2.11) gives the power in the signal $x(t)$: (Average power)

$$P_{av} = \frac{1}{T} \int_T |x(t)|^2 dt$$

(Average power in Watts dissipated by a 1Ω resistor). $\rightarrow$ Normalized to 1Ω .

Power is characterized by root-mean-squared values
RMS.

Example: (HW)
2 Determine the power in the signal:

$$x(t) = A_1 \cos(2\pi f_1 t + \theta_1) + A_2 \cos(4\pi f_1 t + \theta_2)$$

(Hint: use the orthogonality properties above)

If two signals are orthogonal. The

$$P_{x+y} = P_x + P_y$$

where P_x, P_y is the average power in signals $x(t)$ and

$y(t)$ respectively and $x(t)$ — P_{x+y} is the avg. power

in their sum.

→ Power Ratios:

Often the ratio between powers of two signals

is of interest. Two types of power ratios:

(a) Linear → P_2/P_1 (Dimensionless)

(b) Logarithmic → Units: Decibels.

$$10 \log_{10} \frac{P_2}{P_1} = 10 \log_{10} P_2 - 10 \log_{10} P_1$$

Signal to Noise Ratio: The ratio between the power of a signal and the power of the noise (floor) is a fundamental parameter called Signal-to-noise ratio (SNR)

$$\begin{aligned} \text{SNR (dB)} &= 10 \log_{10} \left(\frac{P_{\text{signal}}}{P_{\text{noise}}} \right) \\ &= 10 \log_{10} P_{\text{signal}} - 10 \log_{10} P_{\text{noise}} \end{aligned}$$

An SNR of 0 dB means that noise and signal have equal power.

$\text{SNR} < 0$ (Noise has greater power than the signal)

$\text{SNR} > 0$ (Signal has more power than noise).

Decibel is a relative measure of power.

An absolute measure of power is dBm (relative to

(1mW).

$$\text{power (dBm)} = 10 \log_{10} \left(\frac{P_{\text{mW}}}{1 \text{ mW}} \right)$$

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Another logarithmic unit is dB , power relative to 1W.

$$\text{power (dBW)} = 10 \log_{10} \left(\frac{P_{\text{watts}}}{1 \text{ W}} \right)$$

For example using the above equations

$$1 \text{ W} = 0 \text{ dBW} = 30 \text{ dBm}$$

Remarks:

(a) Communication systems work in a very wide range of power values, (some values are very large while some are very small).

(b) dBm is a very convenient unit for two reasons:

(i) It avoids zeros before and after the decimal point.

(ii) Allows addition to be used instead of multiplication. (logarithmic arithmetic)

Other power measures used are dBm (dB

above reference noise) where the reference level is 1 pW

10^{-12} W. ($0 \text{ dBm} = -90 \text{ dBm} = 1 \text{ pW}$). Another me

is dBmV or dBV. But this requires the actual

level of the impedance R to be known. Standard impedance

values are 75Ω or 50Ω .

Example: Power dBmV: 20 dBmV

$$\Rightarrow V = 10 \text{ mV}$$

$$\text{if } R = 75 \Omega$$

$$\begin{aligned} \text{Power is } \Rightarrow \frac{(10^{-4})}{75} \text{ W} &= 0.00133 \text{ mW} \\ &= -28.75 \text{ dBm} \end{aligned}$$

SECTION 2-3: Fourier Transform

Reading Assignment p. 11 - p. 15.

SECTION 2-4 : Linear Time Invariant Systems

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Reading Assignment.

Keywords:

- (a) Filters
- (b) Low-pass filter
- (c) High-pass filter
- (d) Band-pass filter
- (e) Cutoff frequency
- (f) Stop band
- (g) pass band
- (h) Half power points
- (i) 3-dB Bandwidth.

Quality factor: Characterization of a

bandpass filter:

$$Q = \frac{f_c}{W}$$

$Q \uparrow$: narrow passband

$Q \downarrow$ wide passband.

The frequency response of an ideal low pass filter (22)

with cutoff freq. of ω is:

$$\Pi\left(\frac{f}{2\omega}\right) \equiv \text{graph of } H(f)_{LP}$$

Hw: Derive the impulse response $h_{LP}(t)$ of the ideal lowpass filter using duality.

Hw: Determine the eigenfunctions of LTI systems.

Hint: Example 2-5 (p17)

(From linear Algebra: Eigenvector $\vec{v}$:

$$A\vec{v} = \lambda\vec{v})$$

(From signals and systems:

$$\mathcal{L}\{u(t)\} = A_0 u(t) \quad \text{c/p signal.}$$

$\uparrow$ System $\uparrow$ i/p signal

Section 2-5: Energy and Power Spectral Density:

(23)

ESD + PSD $\rightarrow$ Both rely on a fundamental property
known as Parsavel's Theorem.
Energy Power

HW: Prove that $\langle x(t), y(t) \rangle = \langle X(f), \bar{Y}(f) \rangle$

(Hint: use the definition of inner product and
either (i) FT forward OR (ii) FT inverse.

Therefore energy can be calculated in either time or

frequency: $E_x = \langle x(t), x(t) \rangle = \langle X(f), X(f) \rangle$

$$= \underbrace{\int_{-b}^{+b} |X(f)|^2 df}_{\text{energy}}$$

$\Downarrow$

$$|X(f)|^2 \equiv \text{ESD.}$$

Joules/Hz.

ESD specifies how energy is distributed across the spectrum (frequencies).

The counterpart to ESD is PSD. Let $\underline{x}_T(f)$ be the Fourier Transform of the truncated $x_T(t)$: a snapshot of or segment of $x(t)$ over a duration T . Then according to Parseval's, the total power is the inner power of $\underline{x}_T(f)$ with it self.

$$P_x = \langle \underline{x}_T(f), \underline{x}_T(f) \rangle = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-\infty}^{+\infty} |\underline{x}_T(f)|^2 df$$

Changing the order of integration and limit we have:

$$P_x = \underbrace{\int_{-\infty}^{+\infty} \lim_{T \rightarrow \infty} \frac{1}{T} |\underline{x}_T(f)|^2 df}_{\text{Power}}$$

Thus $\text{PSD} = S_x(f) = \lim_{T \rightarrow \infty} \frac{1}{T} |\underline{x}_T(f)|^2.$

10/8 & (14/2025) Wednesday

(25)

Lecture: Continuation of ESD, PSD (Section 2-5)

Recall for a power signal:

$$P_X = \langle X(f), X(f) \rangle$$

$$= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-\infty}^{+\infty} |X_T(f)|^2 df.$$

$$= \int_{-\infty}^{+\infty} \lim_{T \rightarrow \infty} \frac{|X(f)|^2}{T} df$$

If the integral with respect to frequency is equal to power, then the function inside the integral must be a PSD $S_X(f)$:

$$P_X = \int_{-\infty}^{+\infty} S_X(f) df$$

$$\Rightarrow S_X(f) = \lim_{T \rightarrow \infty} \frac{1}{T} |X_T(f)|^2.$$

Note that for some signals the limit might be infinite

e.g. if $x(t) = A$ then $X_T(f) = AT \text{sinc}(fT)$

which leads to $S_X(0) = \infty$.

Power is conveniently measured in dBm. A common unit for PSD is dBm/Hz. PSD is also expressed in W/Hz , V_{rms}^2/Hz , $V_{rms}/\sqrt{Hz}$ or $dBV/\sqrt{Hz}$. (26)

Two important properties of PSD:

$$(a) S_{\mathbf{x}}(f) \geq 0$$

$$(b) \operatorname{Im}\{S_{\mathbf{x}}(f)\} = 0 \quad (\text{It is real valued})$$

or $S_{\mathbf{x}}(f) \in \mathbb{R}$.

PSD can be specified as a one sided or two sided function.

We have:

- By definition PSD is 2-sided
- Instrumentation unit uses 1-sided PSD.

This leads to some confusion.

When we work with two-sided PSD, negative

(27)

frequencies must be taken into account and the power in the band (f_1, f_2) is

$$P_{f_1, f_2} = \int_{-f_2}^{-f_1} S_X(f) df + \int_{f_1}^{f_2} S_X(f) df.$$

of $A \cos(2\pi f_0 t)$

For example the power in the band (f_1, f_2) that

includes f_0 is

$$P_{f_1, f_2} = \frac{A^2}{4} + \frac{A^2}{4} = \frac{A^2}{2}$$

(The power in the band $[f_1, f_2]$ is the same as the power in the band (f_1, f_2) if and only if $S_X(f)$ does not have a Dirac Delta at exactly f_1 or f_2).

Where there is ambiguity we will denote the one-sided

PSD, $S_X^+(f)$ as:

$$S_X^+(f) = \begin{cases} S_X(0) & \text{if } f=0 \\ 2S_X(f) & f>0 \end{cases}$$

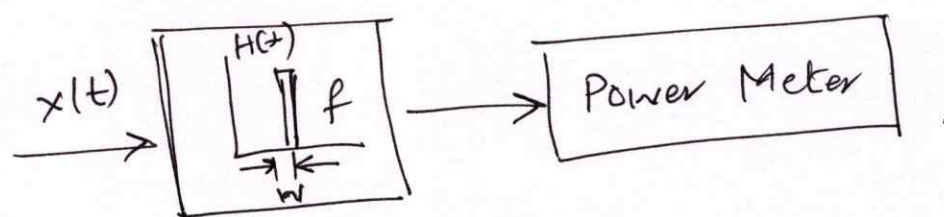
If we use the one-sided PSD, then the integral of the PSD over only the non-negative frequencies accounts for all the power:

$$P_x = \int_0^{\infty} S_{\tilde{x}}^+(f) df$$

and the power in the band (f_1, f_2) is

$$P_{f_1, f_2} = \int_{f_1}^{f_2} S_{\tilde{x}}^+(f) df = \frac{A^2}{2}.$$

Measuring Power: Using power meters.



narrow band filter

w : resolution bandwidth

If w is sufficiently small, the PSD over this bandwidth can be assumed to be a constant equal to

$\frac{P_x}{W}$. The units can be dBm.

$$P_x \text{ (dBm)} = S_x^+ \text{ (dBm/Hz)} + 10 \log_{10} W.$$

HW/Quiz: The average power is measured to be

-60 dBm over a bandwidth of 5 kHz, then the one sided! PSD ?

$$-60 - 10 \log_{10} (5 \times 10^3) = -9 \text{ dBm/Hz}.$$

Quiz: (page 107, 31/458) Example 2-6.
(19) on page)

SECTION 2-6: Bandwidth.

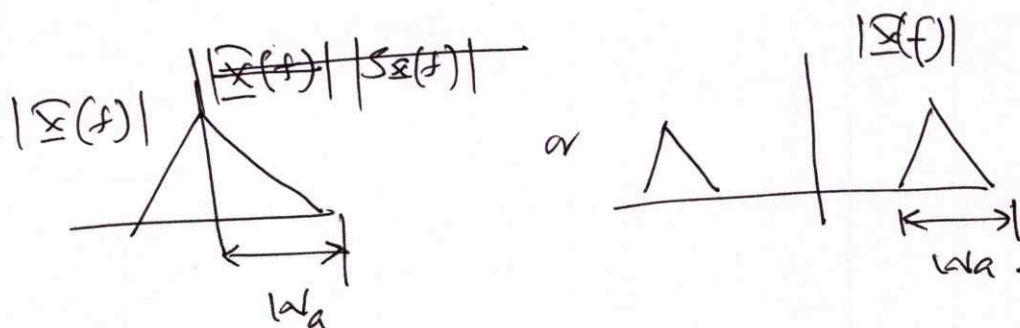
Range of frequencies a signal occupies is called bandwidth;

"bandwidth" $\equiv$ "spectrum occupancy".

Several ways to define BW.

$$\text{Absolute bandwidth: } BW_{abs} = \overline{\{W : S_X(f) > 0\}}$$

For real signals, spectral occupancy is measured only for positive frequencies and the bandwidth is said to be ONE-SIDED, because for real $x(t)$, $\underline{S}(f)$ is even.



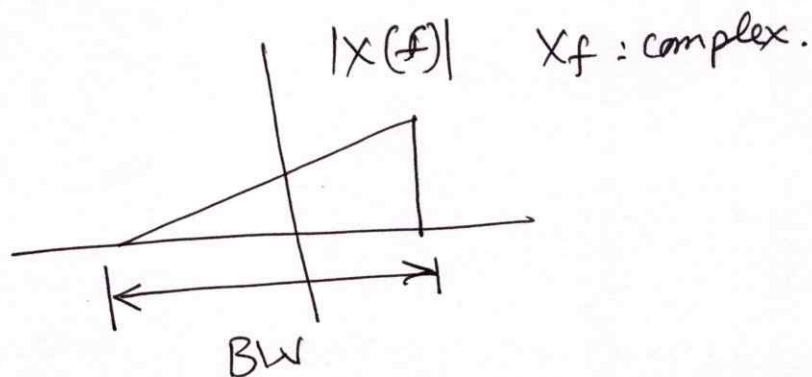
As shown above for real signals- $X(f) = 0$ outside $-W_a \leq f \leq W_a$, the bandwidth is the smallest non-negative frequency W_a such that $\underline{X}(f) = 0$:

$$\underline{S}_X(f) = 0, \quad \forall f \geq |W_a|$$

Such signals are bandlimited.

The spectrum of complex valued signals is not symmetric around 0.

The BW measure in this case referred to as two sided.



Sometimes (Most of the time) $BW_{\text{absolute}} \approx \infty$.

For such cases BW is defined as the range of frequencies for which $S_X(f)$ or $\underline{X}(f)$ is greater than a certain threshold like $\frac{1}{\sqrt{2}} = 0.707$ of the maximum.

This means $\frac{1}{2}$ of the PSD maximum.

3-dB BW $\underline{S_X}(W_{3dB}) = \frac{1}{2} \max_f S_X(f)$

This is called 3-dB bandwidth, because in this

frequency range PSD is within half of the maximum

value and $10 \log_{10} \frac{1}{2} = -3 \text{ dB}$.

More generally, $\bar{X}$ -dB bandwidth is the frequency range over which there is attenuation by most of $\bar{X}$ dB from the maximum. — $\bar{X}$ -dB is useful for characterizing filters.

ESSENTIAL OR FUNCTIONAL BW:

Band across which 95% or 99% energy (or power) resides. The essential band-width W_e is defined for an energy signal as

$$\int_{-W_e}^{W_e} S_{\underline{x}}(f) df = \alpha \int_{-\infty}^{+\infty} S_{\underline{x}}(f) df.$$

Typically $\alpha = 0.9, 0.95$ or 0.99 .

FCC considers fractional power bandwidth as the most appropriate measure of BW.

For power signals, fractional power $< \infty$ even
BW

for signals that have $BW_{absolute} = \infty$.

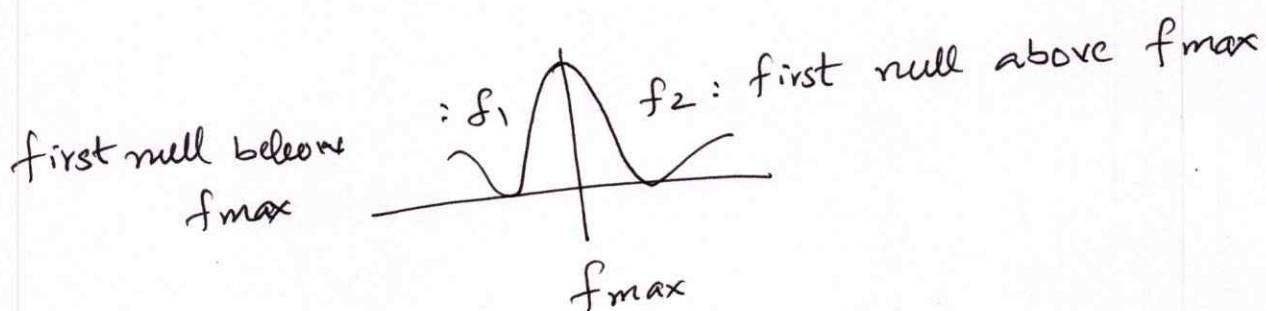
Another measure of bandwidth is called the rms or

Gabor bandwidth

$$W_{rms} = \left[\frac{\int_{-\infty}^{+\infty} f^2 S(f) df}{\int_{-\infty}^{+\infty} S(f) df} \right]^{1/2}$$

W_{rms} can be ∞ and is less useful.

Null-to-Null BW: $f_2 - f_1$;



If $f_{max} = 0$ then null-to-null bandwidth is

a two sided measure even for real valued signal.

The rectangular equivalent BW

Energy signals: BW of a signal that has the same energy as one with a rectangular ESD.

It is the freq. BW such that area under the rectangle is equal to the area under $|X(f)|^2$

It can be defined as:

$$W_{re} = \frac{1}{2 \max_f |X(f)|^2} \int_{-\infty}^{+\infty} |X(f)|^2 df$$

The ~~measure~~^{factor} $\frac{1}{2}$ is present because it's a one-sided measure.

Power Signals: Rect BW. is the BW of the signal with the same power and rectangular PSD. In this case the power is the area of a rectangle. one side of which is W_{rec} . If the max of the one sided PSD is at

a distance freq. f_c , the rectangular equivalent bandwidth of a signal with total power P is:

(35)

$$\text{where } S_x(f) = P$$

~~Exam~~ Q: Example 2-7: Bandwidth of a Rectangular Pulse of duration τ . (page 33/21)

It is seen that in general the BW of a signal is inversely proportional to its duration τ .

$$BW = \frac{\alpha}{\tau}$$

where α depends on the specific definition of BW.

Q: Example 2-8: Rectangular Eq. BW (pp 22 to 23).

Q: Example 2-9: BW of a first order filter